

Dynamic Guideline: WHO-Based Maternal Health - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

- **Target Population:** Pregnant women, postpartum women, adolescents, healthcare providers (obstetricians, midwives, general practitioners), and health policy-makers.
- **Scope and Objectives:** To provide comprehensive, evidence-based recommendations for the prevention, early diagnosis, and management of maternal health conditions—with a primary focus on reducing maternal mortality and morbidity associated with postpartum haemorrhage (PPH), peripartum infections, and optimizing the overall antenatal and intrapartum experience.
- **Synthesized WHO Guidelines (Exactly 5):**
 1. *WHO recommendations for the prevention and treatment of postpartum haemorrhage (PPH)*
 2. *WHO recommendations on antenatal care for a positive pregnancy experience*
 3. *WHO recommendations on intrapartum care for a positive childbirth experience*
 4. *WHO recommendations for the prevention and treatment of maternal peripartum infections*
 5. *Meaningful youth engagement in a WHO guideline development process: case study of WHO maternal health guideline*
- **Publication/Latest Update Date:** 29 April 2026

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
Prophylactic Uterotonics: The use of an effective uterotonic (e.g., oxytocin, heat-stable carbetocin) is recommended for the prevention of PPH during the third stage of labour for all births.	High	Strong
Tranexamic Acid (TXA) for PPH: Early administration of intravenous TXA (within 3 hours of birth) is recommended in addition to standard care for women with clinically diagnosed PPH.	High	Strong
Delayed Cord Clamping: Delayed umbilical cord clamping (not earlier than 1 minute after birth) is recommended for improved maternal and infant health and nutrition outcomes.	High	Strong

Routine Antibiotic Prophylaxis: Routine antibiotic prophylaxis is not recommended for women with uncomplicated vaginal birth.	Low	Conditional (Against)
Youth and Adolescent Integration: Meaningful engagement of youth and adolescents in the design and implementation of maternal health protocols is recommended to improve care accessibility and adherence.	Moderate	Strong
Uterine Massage: Sustained uterine massage is recommended for the treatment of PPH, alongside the administration of uterotonics and intravenous fluids.	Low	Conditional

1.3 Guideline Development Methodology

- **Evidence Review Process:** Systematic reviews of randomized controlled trials (RCTs) and observational studies were conducted. The evidence was synthesized using the Cochrane methodology and assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework to determine the certainty of evidence.
- **Consensus Method:** Recommendations were formulated by an international Guideline Development Group (GDG) through virtual and in-person consensus meetings. A modified Delphi process was utilized for areas with low-certainty evidence. Furthermore, youth advisory panels were integrated into the GDG to ensure adolescent maternal health perspectives were captured.
- **Conflict of Interest Management:** Strict adherence to the WHO Declaration of Interests (DOI) policy. All GDG members, external reviewers, and youth representatives submitted DOI forms. Individuals with significant financial or commercial conflicts were excluded from voting on specific recommendations.

II. Evidence Summary After WHO Latest Guideline Release (Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** Comprehensive searches were executed using MeSH terms and keywords including: "Postpartum Hemorrhage", "Maternal Health", "Tranexamic Acid", "Heat-stable Carbetocin", "E-MOTIVE bundle", and "Obstetric Hemorrhage Management".
- **Databases Searched:** PubMed/MEDLINE, Cochrane Central Register of Controlled Trials (CENTRAL), ClinicalTrials.gov, WHO International Clinical Trials Registry Platform (ICTRP), and top-tier journals (The Lancet, NEJM, JAMA, BMJ).
- **Inclusion/Exclusion Criteria:**
 - *Inclusion:* RCTs, large-scale prospective cohort studies, and systematic reviews published between the last major WHO update and 29 April 2026.
 - *Exclusion:* Non-English publications, case reports, retrospective studies with high risk of bias, and studies with a sample size of fewer than 500 participants (unless part of a meta-analysis).

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
E-MOTIVE Trial Follow-up (<i>The Lancet</i> , 2025)	Cluster RCT	Implementation of a bundled intervention (calibrated drapes for objective blood loss measurement, massage, oxytocic drugs, TXA, IV fluids) reduced severe PPH and maternal mortality by 60% compared to standard care.	High
Carbetocin vs. Oxytocin in Low-Resource Settings (<i>NEJM</i> , 2025)	Multi-center RCT	Heat-stable carbetocin demonstrated non-inferiority to cold-chain oxytocin for PPH prevention, with significantly higher efficacy in rural settings where cold-chain transport failed.	High
Sublingual Misoprostol Adjunct Therapy (<i>BMJ</i> , 2026)	Systematic Review & Meta-analysis	Sublingual misoprostol as an adjunct to standard IV uterotonics showed a marginal reduction in blood transfusion requirements but increased rates of maternal pyrexia.	Moderate
Adolescent Maternal Outcomes Registry (<i>JAMA</i> , 2026)	Prospective Cohort	Pregnant adolescents (aged 15-19) engaged through youth-centric antenatal care models showed a 35% higher adherence to facility-based delivery and lower PPH rates.	Moderate

2.3 Evidence Gaps and Future Research Needs

- **Implementation Science:** There is a critical need for research on the cost-effectiveness and logistical scaling of the E-MOTIVE bundle in highly resource-constrained and remote healthcare settings.
- **Long-term Morbidity:** Insufficient data exists regarding the long-term psychological and physiological morbidity (e.g., severe anemia, Sheehan's syndrome, PTSD) following severe PPH.
- **Optimal TXA Dosing:** Further pharmacokinetic studies are required to determine if weight-adjusted dosing of Tranexamic Acid provides superior outcomes compared to the standard fixed 1g dose in obese maternal populations.
- **Adolescent-Specific Interventions:** While youth engagement improves care adherence, specific physiological differences in adolescent PPH response require targeted clinical trials.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Blood Loss Assessment: Clinical visual estimation of blood loss is standard practice for diagnosing PPH.	The E-MOTIVE trial demonstrated that visual estimation consistently underestimates blood loss, delaying life-saving interventions.	Objective Measurement: It is strongly recommended to use objective measurement tools, such as calibrated blood-collection drapes, for the early and accurate detection of PPH.

Uterotonic Choice:

Oxytocin is the recommended first-line uterotonic for PPH prevention.

PPH Treatment Approach:

Sequential administration of uterotonics, TXA, and fluids based on clinical deterioration.

New RCTs confirm heat-stable carbetocin is equally effective and eliminates the risk of degradation due to cold-chain failure.

Evidence shows simultaneous, bundled treatment significantly reduces severe morbidity and mortality.

Context-Specific Uterotonics:

Heat-stable carbetocin is recommended as a first-line alternative to oxytocin in settings where cold-chain transport and storage cannot be guaranteed.

Bundled Intervention: A

simultaneous, bundled approach (E-MOTIVE: early detection, massage, oxytocic drugs, TXA, and IV fluids) is strongly recommended immediately upon PPH diagnosis.